

Integration of Basic Trigonometric Functions continued

33. Evaluate the definite integrals:

(g) $\int_0^{\frac{\pi}{4}} (\cos x - \sin x) dx$ (h) $\int_{\frac{\pi}{3}}^{\pi} 1 + \sin 2x dx$ (i) $\int_{-\frac{\pi}{2}}^{\frac{3\pi}{2}} \left(3 \sin 2x - \cos \frac{x}{2} \right) dx$

(j) $\int_0^{\frac{\pi}{3}} \frac{1 + \cos^3 x}{\cos^2 x} dx$ (k) $\int_{\frac{\pi}{4}}^{\frac{\pi}{3}} \frac{\sec^2 x}{\tan^3 x} dx$ (l) $\int_0^{\frac{\pi}{2}} \sqrt{\sin x} \cos x dx$

(g) $\int_0^{\frac{\pi}{4}} (\cos x - \sin x) dx = [\sin x + \cos x]_0^{\frac{\pi}{4}} = \left(\frac{\sqrt{2}}{2} + \frac{\sqrt{2}}{2} \right) - (0 + 1) = \sqrt{2} - 1$

(h) $\int_{\frac{\pi}{3}}^{\pi} (1 + \sin 2x) dx = \left[x - \frac{\cos 2x}{2} \right]_{\frac{\pi}{3}}^{\pi} = \left(\pi - \frac{1}{2} \right) - \left(\frac{\pi}{3} - \frac{1}{2} \left(-\frac{1}{2} \right) \right)$
 $= \frac{2\pi}{3} - \frac{1}{2} - \frac{1}{4} = \frac{2\pi}{3} - \frac{3}{4}$

(i) $\int_{-\frac{\pi}{2}}^{\frac{3\pi}{2}} \left(3 \sin 2x - \cos \frac{x}{2} \right) dx = \left[-\frac{3 \cos 2x}{2} - 2 \sin \frac{x}{2} \right]_{-\frac{\pi}{2}}^{\frac{3\pi}{2}} = \left[-\frac{3}{2} (-1) - 2 \left(\frac{1}{2} \right) \right] - \left[-\frac{3}{2} (-1) - 2 \left(-\frac{\sqrt{2}}{2} \right) \right]$
 $= \frac{3}{2} - \sqrt{2} - \frac{3}{2} - \sqrt{2} = -2\sqrt{2}$

34. Given that $\frac{d^2y}{dx^2} = -12 \cos 2x - 16 \sin 2x$ and when $x=0$, $\frac{dy}{dx} = 8$ and $y=8$, find y in terms of x .

$$\frac{d^2y}{dx^2} = -12 \cos 2x - 16 \sin 2x$$

Integrate term on both sides

$$\frac{dy}{dx} = -12 \frac{\sin 2x}{2} + 16 \frac{\cos 2x}{2} + C_1$$

$$\frac{dy}{dx} = -6 \sin 2x + 8 \cos 2x + C_1$$

when $x=0$, $\frac{dy}{dx} = 8$

$$8 = -6(0) + 8(1) + C_1$$

$$\Rightarrow C_1 = 0$$

$$\therefore \frac{dy}{dx} = -6 \sin 2x + 8 \cos 2x$$

Integrate again on both sides
 $y = -6 \left(-\frac{\cos 2x}{2} \right) + 8 \left(\frac{\sin 2x}{2} \right) + C_2$

$$y = 3 \cos 2x + 4 \sin 2x + C_2$$

when $x=0$, $y=8$

$$8 = 3(1) + 4(0) + C_2$$

$$C_2 = 5$$

$$\therefore y = 3 \cos 2x + 4 \sin 2x + 5$$

33

$$\begin{aligned}
 (j) \quad \int_0^{\pi/3} \frac{1 + \cos^3 x}{\cos^2 x} dx &= \int_0^{\pi/3} \left(\frac{1}{\cos^2 x} + \frac{\cos^3 x}{\cos^2 x} \right) dx = \int_0^{\pi/3} (\sec^2 x + \cos x) dx \\
 &= \left[\tan x + \sin x \right]_0^{\pi/3} = \left(\sqrt{3} + \frac{1}{2} \right) - (0) \\
 &= \frac{2\sqrt{3}}{2}
 \end{aligned}$$

$$\begin{aligned}
 (k) \quad \int_{\pi/4}^{\pi/3} \frac{\sec^2 x}{\tan^3 x} dx &= \int_{\pi/4}^{\pi/3} \sec^2 x \cdot (\tan x)^{-3} dx = \left[\frac{(\tan x)^{-2}}{-2} \right]_{\pi/4}^{\pi/3} \\
 &= \frac{1}{-2} \left[\frac{1}{\tan^2 x} \right]_{\pi/4}^{\pi/3} \\
 &= \frac{1}{-2} \left[\frac{1}{3} - \frac{1}{1} \right] \\
 &= \frac{1}{-2} \left[\frac{-2}{3} \right] = \frac{1}{3}
 \end{aligned}$$

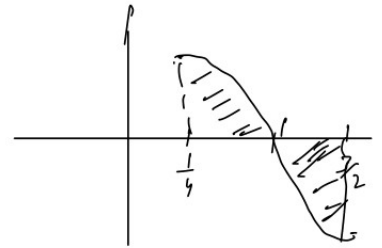
$$\begin{aligned}
 (l) \quad \int_0^{\pi/2} \sqrt{\sin x} \cos x dx &= \int_0^{\pi/2} (\cos x) (\sin x)^{1/2} dx = \left[\frac{(\sin x)^{3/2}}{3/2} \right]_0^{\pi/2} \\
 &= \frac{2}{3} [1 - 0] = \frac{2}{3} \checkmark
 \end{aligned}$$

Integration of Basic Trigonometric Functions continued

35. Find the area bounded by the curve $y = \sin \pi x$ and the x -axis, between $x = \frac{1}{4}$ and

$$x = \frac{3}{2}.$$

$$\begin{aligned} \text{Area } A &= \int_{\frac{1}{4}}^1 \sin \pi x \, dx + \left| \int_1^{\frac{3}{2}} \sin \pi x \, dx \right| \\ &= -\left[\frac{\cos \pi x}{\pi} \right]_{\frac{1}{4}}^1 + \left| -\left[\frac{\cos \pi x}{\pi} \right]_1^{\frac{3}{2}} \right| \\ &= \frac{-1}{\pi} \left(-1 - \frac{1}{\sqrt{2}} \right) + \left| \frac{-1}{\pi} [0 - (-1)] \right| \\ &= \frac{1}{\pi} + \frac{\sqrt{2}}{2\pi} + \frac{1}{\pi} = \frac{4 + \sqrt{2}}{2\pi} \text{ units}^2 \end{aligned}$$



36. The region bounded by the curve $y = \tan 2x$ and the x -axis, from $x = 0$ to $x = \frac{\pi}{6}$ is rotated about the x -axis to form a solid.

(a) Show that the volume of the solid is given by $V = \pi \int_0^{\frac{\pi}{6}} (\sec^2 2x - 1) dx$.

(b) Find the exact volume of the solid.

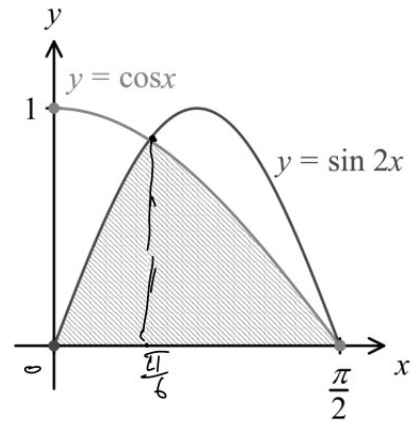
(a) Volume, $V_k = \pi \int_0^{\frac{\pi}{6}} (\tan^2 2x) dx = \pi \int_0^{\frac{\pi}{6}} (\sec^2 2x - 1) dx$

(b)
$$\begin{aligned} V_k &= \pi \left[\frac{\tan 2x}{2} - x \right]_0^{\frac{\pi}{6}} \\ &= \pi \left[\frac{\sqrt{3}}{2} - \frac{\pi}{6} \right] \text{ units}^3 \end{aligned}$$

Integration of Basic Trigonometric Functions continued

37. The diagram shows the graphs of the functions $y = \cos x$ and $y = \sin 2x$ between $x = 0$ and $x = \frac{\pi}{2}$.

Calculate the area of the shaded region.



$$\begin{aligned} \cos x &= \sin 2x \\ \cos x &= 2 \sin x \cos x \\ \cos x (1 - 2 \sin x) &= 0 \\ \cos x &= 0 & 2 \sin x &= 1 \\ x &= \frac{\pi}{2} & \sin x &= \frac{1}{2} \\ & & x &= \frac{\pi}{6} \end{aligned}$$

$$\begin{aligned} \text{Area, } A &= \int_0^{\pi/6} \sin 2x \, dx + \int_{\pi/6}^{\pi/2} \cos x \, dx \\ &= -\left(\frac{\cos 2x}{2}\right)_0^{\pi/6} + \left[\sin x\right]_{\pi/6}^{\pi/2} \\ &= -\frac{1}{2}\left[\frac{1}{2} - 1\right] + \left[1 - \frac{1}{2}\right] \\ &= \frac{1}{4} + \frac{1}{2} \\ &= \frac{3}{4} \text{ units}^2 \end{aligned}$$

Further Calculus of Trigonometric Functions

38. Find the following:

(a) $\int \cos^2 3x \, dx$

(b) $\int 3 \sin^2 \frac{x}{2} \, dx$

(c) $\int 2 - 3 \sin^2 x \, dx$

(d) $\int (1 + \cos x)^2 \, dx$

(e) $\int \sin^2(2x-1) \, dx$

(f) $\int \frac{\cos(2x-1)}{\sin^2(2x-1)} \, dx$

(g) $\int (\cos^4 x - \sin^4 x) \, dx$

(h) $\int 2 \sin x \cos 2x \, dx$

(i) $\int \frac{1 + \sin^2 x}{\cos^2 x} \, dx$

(a) $\int \cos^2 3x \, dx = \int \left(\frac{1 + \cos 6x}{2} \right) dx = \frac{1}{2} x + \frac{\sin 6x}{12} + C$

(b) $\int 3 \sin^2 \frac{x}{2} \, dx = 3 \int \left(\frac{1 - \cos x}{2} \right) dx = \frac{3}{2} \left[x - \sin x \right] + C$

(c) $\int (2 - 3 \sin^2 x) \, dx = \int \left(2 - 3 \left(\frac{1 - \cos 2x}{2} \right) \right) dx = \int \left(2 - \frac{3}{2} + \frac{3}{2} \cos 2x \right) dx = \int \left(\frac{1}{2} + \frac{3}{2} \cos 2x \right) dx = \frac{x}{2} + \frac{3 \sin 2x}{4} + C$

(d) $\int (1 + \cos x)^2 \, dx = \int (1 + 2 \cos x + \cos^2 x) \, dx = \int \left(1 + 2 \cos x + \frac{1 + \cos 2x}{2} \right) dx = \int \left(\frac{3}{2} + 2 \cos x + \frac{1}{2} \cos 2x \right) dx = \frac{3x}{2} + 2 \sin x + \frac{\sin 2x}{4} + C$

(e) $\int \sin^2(2x-1) \, dx = \int \left(\frac{1 - \cos 2(2x-1)}{2} \right) dx = \frac{1}{2} \int (1 - \cos(4x-2)) \, dx = \frac{1}{2} \left[x - \frac{\sin(4x-2)}{4} \right] + C$

(f) $\int \frac{\cos(2x-1)}{\sin^2(2x-1)} \, dx = \frac{1}{2} \int 2 \cos(2x-1) \cdot \left[\sin(2x-1) \right]^{-2} dx = \frac{1}{2} \cdot \frac{(\sin(2x-1))^{-1}}{-1} + C = \frac{-1}{2 \sin(2x-1)} + C$

(g) $\int (\cos^4 x - \sin^4 x) \, dx = \int (\cos^2 x - \sin^2 x)(\cos^2 x + \sin^2 x) \, dx = \int \cos 2x \, dx = \frac{\sin 2x}{2} + C$

(h) $\int 2 \sin x \cos 2x \, dx = \int 2 \sin x (2 \cos^2 x - 1) \, dx = \int (4 \sin x \cos^2 x - 2 \sin x) \, dx = -4 \int (\sin x) \cos^2 x \, dx - 2 \int \sin x \, dx = -4 \frac{\cos^3 x}{3} - 2(-\cos x) + C = \frac{-4}{3} \cos^3 x + 2 \cos x + C$

(i) $\int \frac{1 + \sin^2 x}{\cos^2 x} \, dx = \int (\sec^2 x + \tan^2 x) \, dx = \int \sec^2 x + (\sec^2 x - 1) \, dx = \int (2 \sec^2 x - 1) \, dx = 2 \tan x - x + C$

Further Calculus of Trigonometric Functions continued

39. Evaluate the following:

(a) $\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \cos^2 x \, dx$

(b) $\int_0^{\frac{\pi}{2}} \sin^2 x \cos x \, dx$

(c) $\int_{\frac{\pi}{6}}^{\frac{\pi}{3}} (\sin x + \cos x)^2 \, dx$

(d) $\int_{\frac{\pi}{3}}^{\frac{5\pi}{6}} \frac{\sec^2 x}{(\tan x - 1)^2} \, dx$

(a) $\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \cos^2 x \, dx = \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{1 + \cos 2x}{2} \, dx = \frac{1}{2} \left[x + \frac{\sin 2x}{2} \right]_{-\frac{\pi}{2}}^{\frac{\pi}{2}} = \frac{1}{2} \left[\left(\frac{\pi}{2} + 0 \right) - \left(-\frac{\pi}{2} + 0 \right) \right] = \frac{\pi}{2}$

(b) $\int_0^{\frac{\pi}{2}} \sin^2 x \cos x \, dx = \int_0^{\frac{\pi}{2}} \cos x \cdot (\sin x)^2 \, dx = \left[\frac{(\sin x)^3}{3} \right]_0^{\frac{\pi}{2}} = \frac{1}{3} - 0 = \frac{1}{3}$

(c) $\int_{\frac{\pi}{6}}^{\frac{\pi}{3}} (\sin x + \cos x)^2 \, dx = \int_{\frac{\pi}{6}}^{\frac{\pi}{3}} (\sin^2 x + 2 \sin x \cos x + \cos^2 x) \, dx = \int_{\frac{\pi}{6}}^{\frac{\pi}{3}} (1 + \sin 2x) \, dx = \left[x - \frac{\cos 2x}{2} \right]_{\frac{\pi}{6}}^{\frac{\pi}{3}} = \left(\frac{\pi}{3} - \left(-\frac{1}{4} \right) \right) - \left(\frac{\pi}{6} - \frac{1}{4} \right) = \frac{\pi}{6} + \frac{1}{2}$

(d) $\int_{\frac{\pi}{3}}^{\frac{5\pi}{6}} \frac{\sec^2 x}{(\tan x - 1)^2} \, dx = \int_{\frac{\pi}{3}}^{\frac{5\pi}{6}} \sec^2 x \cdot (\tan x - 1)^{-2} \, dx = \left[\frac{(\tan x - 1)^{-1}}{-1} \right]_{\frac{\pi}{3}}^{\frac{5\pi}{6}} = - \left[\frac{1}{\tan x - 1} \right]_{\frac{\pi}{3}}^{\frac{5\pi}{6}} = - \left[\frac{1}{\left(-\frac{1}{\sqrt{3}} \right) - 1} - \frac{1}{\sqrt{3} - 1} \right] = - \left[\frac{\sqrt{3}}{1 - \sqrt{3}} - \frac{1}{\sqrt{3} - 1} \right] = \frac{\sqrt{3}}{\sqrt{3} + 1} + \frac{1}{\sqrt{3} - 1} = \frac{3 - \sqrt{3} + \sqrt{3} + 1}{(\sqrt{3} + 1)(\sqrt{3} - 1)} = \frac{4}{2} = 2$

Further Calculus of Trigonometric Functions continued

40. Given the function $y = \sin^2 x$ for $-\pi \leq x \leq \pi$,

- find the coordinates of the stationary points and determine their nature;
- find the coordinates of the points of inflection;
- find any intercepts on the coordinate axes;
- sketch the graph of $y = \sin^2 x$;
- find the area bounded by the curve, the x-axis, and the ordinates $x = 0$, $x = \pi$.

(a) $y = \sin^2 x$

$$y' = 2 \sin x \cos x$$

$$y' = 0$$

$$\sin 2x = 0, \quad -\pi \leq x \leq \pi$$

$$-2\pi \leq 2x \leq 2\pi$$

$$2x = \pm 2\pi, \pm \pi, 0$$

$$x = \pm \pi, \pm \frac{\pi}{2}, 0$$

Stationary points are $(-\pi, 0), (\pi, 0), (-\frac{\pi}{2}, 1), (\frac{\pi}{2}, 1), (0, 0)$

$$y' = \sin 2x$$

$$y'' = 2 \cos 2x$$

$$x = -\pi, y'' = 2 \cos 2(-\pi) = 2 \cos 2\pi = 2(1) > 0 \rightarrow \text{min at } (-\pi, 0)$$

$$x = \pi, y'' = 2 \cos 2\pi = 2 > 0 \rightarrow \text{min at } (\pi, 0)$$

$$x = -\frac{\pi}{2}, y'' = 2 \cos 2(-\frac{\pi}{2}) = 2 \cos(-\pi) = 2 \cos \pi = -2 < 0 \rightarrow \text{max at } (-\frac{\pi}{2}, 1)$$

$$x = \frac{\pi}{2}, y'' = 2 \cos 2(\frac{\pi}{2}) = 2 \cos(\pi) = -2 < 0 \rightarrow \text{max at } (\frac{\pi}{2}, 1)$$

$$x = 0, y'' = 2 \cos 2(0) = 2 > 0 \rightarrow \text{min at } (0, 0)$$

Max at $(\pm \frac{\pi}{2}, 1)$, Min at $(0, 0), (\pm \pi, 0)$

(b) $y'' = 2 \cos 2x, \quad -2\pi \leq 2x \leq 2\pi$

$$2 \cos 2x = 0$$

$$\cos 2x = 0$$

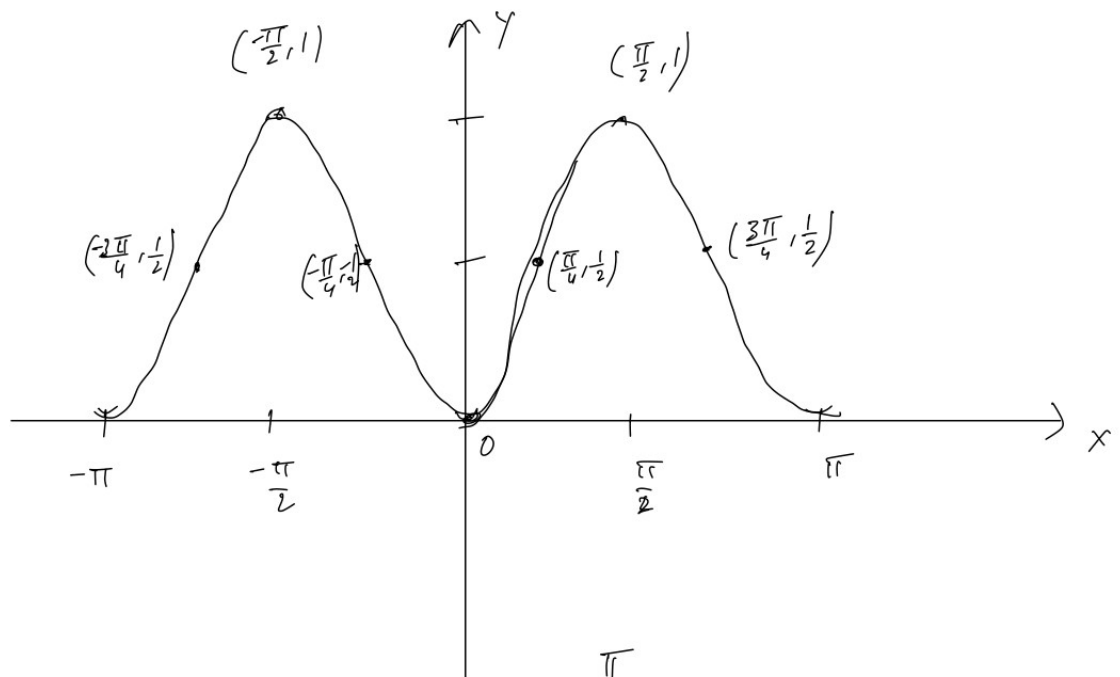
$$2x = \pm \frac{\pi}{2}, \pm \frac{3\pi}{2}$$

$$x = \pm \frac{\pi}{4}, \pm \frac{3\pi}{4}$$

x	$-\pi$	$-\frac{3\pi}{4}$	$-\frac{\pi}{2}$	$-\frac{\pi}{4}$	0	$\frac{\pi}{4}$	$\frac{\pi}{2}$	$\frac{3\pi}{4}$	π
y'	+	0	-	0	+	0	-	0	+

concave down

d

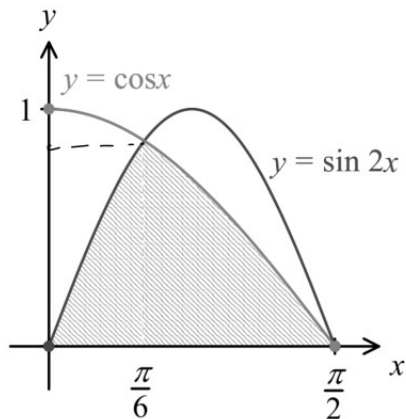


e

$$\begin{aligned}
 \text{Area, } A &= \int_0^{\pi} \sin^2 x \, dx = \int_0^{\pi} \frac{1 - \cos 2x}{2} \, dx \\
 &= \frac{1}{2} \left[x - \frac{\sin 2x}{2} \right]_0^{\pi} \\
 &= \frac{1}{2} \left[(\pi - 0) - (0 - 0) \right] \\
 &= \frac{\pi}{2} \text{ units}^2
 \end{aligned}$$

Further Calculus of Trigonometric Functions continued

41.



The diagram shows the graphs of the functions $y = \cos x$ and $y = \sin 2x$ between $x = 0$ and

$$x = \frac{\pi}{2}.$$

The shaded region is rotated about the x -axis to form a solid of revolution. Find the volume of this solid, giving your answer correct to 3 decimal places.

$$\begin{aligned}
 V_k &= \pi \int_0^{\pi/6} \sin^2 x \, dx + \pi \int_{\pi/6}^{\pi/2} \cos^2 x \, dx \\
 &= \pi \int_0^{\pi/6} \frac{1 - \cos 4x}{2} \, dx + \pi \int_{\pi/6}^{\pi/2} \frac{1 + \cos 2x}{2} \, dx \\
 &= \frac{\pi}{2} \left[x - \frac{\sin 4x}{4} \right]_0^{\pi/6} + \frac{\pi}{2} \left[x + \frac{\sin 2x}{2} \right]_{\pi/6}^{\pi/2} \\
 &= \frac{\pi}{2} \left[\left(\frac{\pi}{6} - \frac{1}{4} \cdot \frac{\sqrt{3}}{2} \right) - (0) \right] + \frac{\pi}{2} \left[\left(\frac{\pi}{2} + 0 \right) - \left(\frac{\pi}{6} + \frac{1}{2} \cdot \frac{\sqrt{3}}{2} \right) \right] \\
 &= \frac{\pi}{2} \left[\frac{\pi}{6} - \frac{\sqrt{3}}{8} \right] + \frac{\pi}{2} \left[\frac{\pi}{3} - \frac{\sqrt{3}}{4} \right] \\
 &= \frac{\pi^2}{12} - \frac{\sqrt{3}\pi}{16} + \frac{\pi^2}{6} - \frac{\sqrt{3}\pi}{8} \\
 &= \frac{\pi^2 + 2\pi^2}{12} - \frac{(\sqrt{3}\pi + 2\sqrt{3}\pi)}{16} \\
 &= \frac{\pi^2}{4} - \frac{3\sqrt{3}\pi}{16} \\
 &\approx 1.447 \text{ units}^3
 \end{aligned}$$

Rates of Change

42. The number N of bacteria in a culture is given by the formula $N = 2t^4 - t^2 + 2000$, where t is the time in hours.

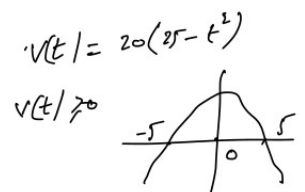
- (a) Find the initial number of bacteria.
 (b) Find the number of bacteria after 5 hours.
 (c) Find the average rate of increase in bacteria in the first 5 hours.
 (d) Find the instantaneous rate of increase in bacteria after 5 hours.

(a) $t=0, N(0) = 2000$
 (b) $t=5, N(5) = 2(5)^4 - 5^2 + 2000 = 3225$ bacteria
 (c) Average rate $= \frac{N(5) - N(0)}{5 - 0} = \frac{3225 - 2000}{5} = 245$ bacteria/hour
 (d) $N'(t) = 8t^3 - 2t$
 $t=5, N'(5) = 8(5)^3 - 2(5) = 990$ bacteria/hour

43. A 500 litre tank is being emptied of water so that after t minutes, the volume of water in the tank is given by $V = a + bt^2$ where a and b are constants.

- (a) If the tank is initially full of water and takes 5 minutes to empty, find the values of a and b .
 (b) Find the rate at which the volume of water is changing :
 (i) when $t = 2.5$ (ii) when the tank is half empty.

(a) $V(t) = a + bt^2$
 $t=0, V(0) = 500$ litre
 $a + b(0)^2 = 500$
 $a = 500$
 $V(5) = 0$
 $a + b(5)^2 = 0$
 $500 + 25b = 0$
 $25b = -500$
 $b = -20$
 $\therefore V(t) = 500 - 20t^2, 0 \leq t \leq 5$



(b) (i) $V'(t) = -40t$
 $V'(2.5) = -40(2.5) = -100$ L/min

(ii) $V(t) = 250$
 $500 - 20t^2 = 250$
 $250 = 20t^2$
 $t^2 = \frac{25}{2} \Rightarrow t = \frac{5}{\sqrt{2}}$
 $V'\left(\frac{5}{\sqrt{2}}\right) = -40\left(\frac{5}{\sqrt{2}}\right) = -100\sqrt{2}$ L/min